

Week 3 Day 5 Capstone — Full AFL Assistant

Executive Report

Product goal. Ship a domain-locked AFL conversational assistant combining exact real-dataset retrieval, match/player prediction, multi-turn memory, guardrails, API/UI access, evaluation, and monitoring. The system fails safely: off-topic and prompt-injection requests are refused, missing statistics are never guessed, and predictions use the consistent language “predicted probability, not a certainty.”

Architecture. LangGraph routes each request through deterministic scope and intent checks. Retrieval uses pandas tools over the supplied real AFL CSVs for player season statistics, round-by-round form, team form, and head-to-head records. Prediction calls the trained match-winner/top-player models. Gemini is used only to phrase verified tool results; deterministic formatting remains available if Gemini is unavailable or rate-limited. Conversation state is keyed by conversation_id. FastAPI exposes /chat and /health, and Streamlit provides a minimal demo UI.

System hardening. Scope checks occur before tool execution. Known prompt-injection patterns such as instruction overrides, developer-mode requests, and persona switching are rejected. Per-conversation rate limiting is applied. Retrieval/prediction exceptions become safe clarification messages. API logging records query, conversation ID, latency, tools called, and prediction metadata. A regression fix now recognizes real AFL player names as in-scope even when no AFL keyword is present. Team extraction preserves the user’s textual order and self-match requests are rejected instead of inventing an opponent.

Evaluation. The combined suite contains 33 cases: factual/prediction behavior, scope guardrails, 3 explicit prompt-injection attempts, and multi-turn coherence. Three additional prediction-sanity regression cases verify team-order preservation and self-match rejection.

Category	Cases	Pass rate
Factual/Prediction	15	100.0%
Multi-turn coherence	5	100.0%
Prompt injection	3	100.0%
Scope	10	100.0%

Weakest category. No category has a lower pass rate in this corrected regression suite; all four categories are at 100%. The highest residual risk area is **scope/abuse handling** because novel adversarial wording can evade a finite rule list. Concrete improvement: expand the adversarial suite with randomized paraphrases and add a model-assisted secondary scope check behind the deterministic guardrail.

Benchmark context. On the genuine 2025 held-out season (215 non-draw games), the leakage-safe match model achieved **66.5%** accuracy, versus **65.6%** for a ladder-position naive benchmark derived chronologically from the same match results, and **55.8%** for a home-team-always baseline. The small lift over the ladder baseline shows that the model should be presented as contextual probabilistic evidence, not certainty.

Known limitations. The prediction model is historical-form based and does not ingest injuries, confirmed lineups, weather, betting markets, or a live fixture feed. “Current” predictions therefore depend on the latest dated records in the supplied CSVs. Gemini free-tier quotas can limit live phrasing. Deterministic guardrails reduce known leakage but cannot guarantee protection against every novel adversarial formulation.

Limitations, Operations & Recommended Next Steps

Deployment. The FastAPI endpoint accepts a message and conversation_id and returns response, prediction metadata, latency, and trace information. Streamlit provides a lightweight demonstration surface. Structured logging establishes a monitoring foundation.

Monitoring checklist. Track response latency (alert when p95 exceeds 5 s), tool error rate (alert above 5%), off-topic leak rate (alert above 1% on the regression suite), prediction accuracy/Brier score drift after each completed round, Gemini/API quota errors, and unusual repeated probing from a conversation. Review logs weekly and immediately after a major data/model change.

Weekly refresh loop. 1) ingest newly completed match results; 2) validate schema, duplicates, dates, teams, and scores; 3) append to the feature table; 4) recompute leakage-safe rolling features; 5) run the 33-case evaluation plus a fresh time-based holdout; 6) retrain only when the new labeled sample is sufficient; 7) compare against the ladder baseline; 8) promote only if quality gates pass, otherwise keep the previous model; 9) version the data/model and record the evaluation metrics.

Recommended next steps. Add a versioned live fixture and ladder feed; add calibration/reliability metrics for probabilities; expand adversarial paraphrase tests; enforce hard tool timeouts around external calls; add model/data version IDs to logs; and maintain rollback artifacts for every retrained model.

Conclusion. The capstone is a complete product-shaped AFL assistant: guarded routing, grounded retrieval, probabilistic predictions, memory, API/UI access, measurable evaluation, benchmark context, and an operational maintenance plan. The corrected implementation specifically prevents player-name over-refusal, preserves prediction team order, rejects self-match requests, derives a real ladder benchmark from match results, and explicitly identifies the remaining highest-risk area.